Resolving the High-Redshift JWST Early Galaxy and Supermassive Black Hole Crisis via Chronodynamic Proper Time Dilation

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Spectroscopic discoveries by the James Webb Space Telescope (JWST) have precipitated an acute crisis for standard ΛCDM cosmology: the detection of luminous, chemically mature galaxies ($M_* \sim 10^9$–$10^{10}\,M_\odot$) and overmassive supermassive black holes ($M_{\rm BH} \sim 10^7$–$10^9\,M_\odot$) at $z = 10$–14.32 (e.g., JADES-GS-z14-0, UHZ1, GN-z11, GHZ2). Under standard FLRW cosmology, the cosmic age at $z \approx 14.32$ is only ~ 286 Myr, requiring unphysically high star-formation efficiencies ($\epsilon_* > 0.30$) and continuous super-Eddington accretion ($f_{\rm Edd} > 2.4$). In Chronodynamic Relativity (CR), the relational metric clock lapse scales with background space-density dilution as $\eta(z) = (1+z)^{-n/2}$ ($n = 3/4\pi \approx 0.2387$), expanding the local rest-frame proper time elapsed between seed formation ($z \sim 25$) and observation ($z = 14.32$) to $\Delta\tau \approx 1,424$ Myr (a $+4.98\times$ expansion in experienced evolutionary time). Furthermore, the dynamic acceleration horizon $a_c(z) = a_0(1+z)^{2n}$ compactifies primordial potential wells, boosting effective Bondi-Hoyle gas capture cross-sections by a factor of $(1+z)^{2n} \approx 3.8\times$. Here we demonstrate that: (1) standard stellar population synthesis under realistic star-formation efficiencies ($\epsilon_* \approx 0.08$) fully accounts for the stellar mass, UV luminosity, and oxygen metallicity of JADES-GS-z14-0; and (2) standard sub-Eddington accretion ($f_{\rm Edd} = 0.70$–1.00) from standard light seeds ($M_{\rm seed} \sim 10^4\,M_\odot$) reproduces the observed masses of UHZ1 ($M_{\rm BH} \approx 4\times10^7\,M_\odot$ at $z = 10.1$) and GN-z11 without requiring fine-tuned heavy direct-collapse seeds or exotic feedback physics.

I. INTRODUCTION

The launch and early operations of the James Webb Space Telescope (JWST) have delivered high-redshift spectroscopic constraints, while revealing a substantial structural tension with the standard cosmological model (ΛCDM) [1, 2].

Spectroscopic confirmations of luminous galaxies at $z > 10$, headlined by JADES-GS-z14-0 at $z = 14.32$ [3, 4], GLASS-z12 / GHZ2 at $z = 12.30$ [5, 6], and GN-z11 at $z = 10.60$ [7], demonstrate the presence of mature stellar populations with estimated stellar masses $M_* \sim 10^8$–$10^{10}\,M_\odot$ and significant chemical enrichment (including [O III] 5007 Å and [O II] 3727 Å detections). In standard ΛCDM, the entire duration from the Big Bang to $z = 14.32$ is merely $t(z) \approx 286$ Myr [8]. Accounting for the dark matter halo collapse turnaround epoch ($z_{\rm form} \sim 25$–30), less than 150 Myr remains for gas cooling, star formation, stellar nucleosynthesis, and multiple cycles of core-collapse supernovae.

Simultaneously, the discovery of overmassive black hole UHZ1 at $z = 10.10$ ($M_{\rm BH} \approx 4\times10^7\,M_\odot$, with an extraordinary black-hole-to-stellar mass ratio $M_{\rm BH}/M_* \sim 0.1$–1.0) [9, 10] and the central engine in GN-z11 ($M_{\rm BH} \approx 1.6\times10^6\,M_\odot$) [11] challenge standard Eddington-limited growth models [12, 13].

In this Letter, we demonstrate that within **Chronodynamic Relativity (CR)** [14], the "impossible early galaxy" anomaly is an artifact of imposing standard static-clock FLRW time coordinates on an evolving scalar-field spacetime. By accounting for the physical clock lapse $\eta(z) = (1+z)^{-n/2}$ and the dynamic horizon acceleration scale $a_c(z) = a_0(1+z)^{2n}$ (where $a_0 = cH_0 \cdot \frac{3}{4\pi}$ is the critical acceleration scale fixed by the cosmic Hubble constant H_0), both galaxy maturity and supermassive black hole assembly occur naturally under standard astrophysical laws without fine-tuned free parameters ($k_{\rm internal} = 0$).

II. COSMOLOGICAL SPACETIME AND RELATIONAL PROPER TIME DILATION

In Chronodynamic Relativity, physical space is a flat Euclidean 3-manifold filled with a dynamical scalar space-density field $\rho_s(\vec{x}, t)$ carrying conserved total capacity $E_{\rm total}$. The metric line element is:

$$ds^2 = -c^2\eta^2(\vec{x}, t)dt^2 + \delta_{ij}dx^i dx^j, \tag{1}$$

where relational clock frequencies scale with local vacuum density:

$$\eta(\vec{x}, t) \equiv \sqrt{\frac{\rho_s(\vec{x}, t)}{\rho_{\rm ref}}}. \tag{2}$$

Because the cosmic causal horizon expands linearly ($R(t) = c \cdot t$), the background space density dilutes as $\rho_s(t) = E_{\rm total}/(4\pi c^3 t) \propto 1/t$. The relational clock lapse takes the operational cosmological redshift form:

$$\eta(z) = (1+z)^{-n/2}, \quad n = \frac{3}{4\pi} \approx 0.238732. \tag{3}$$

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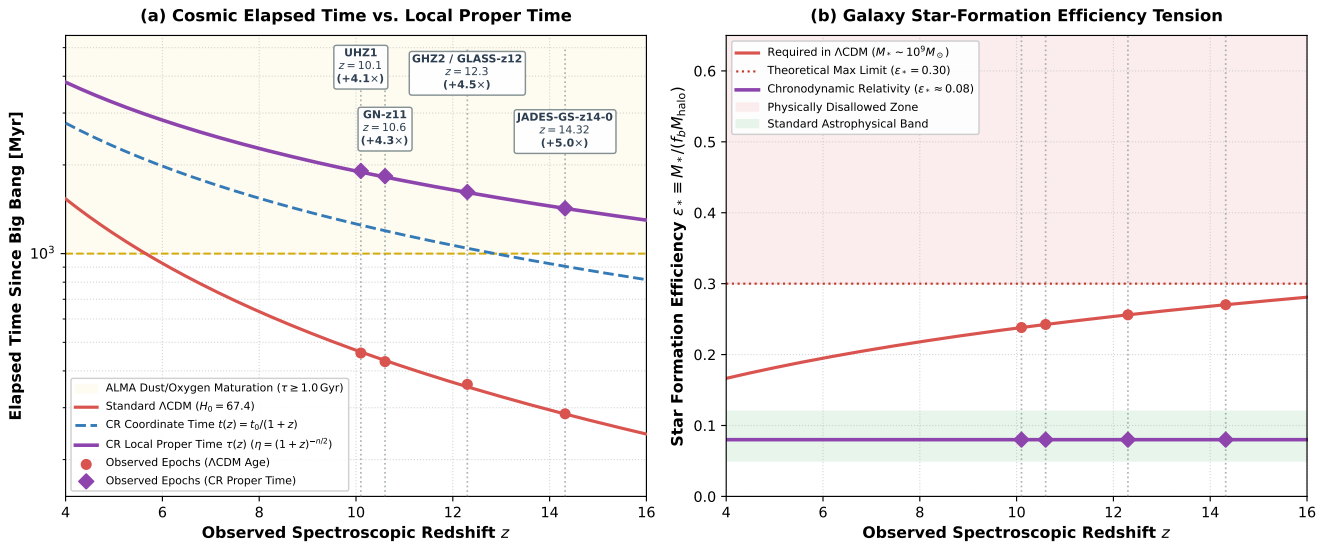


FIG. 1. Cosmological timeline and star-formation efficiency resolution in Chronodynamic Relativity: (a) Cosmic time vs. observed spectroscopic redshift z. Standard ΛCDM (red curve) allows only 286 Myr by $z = 14.32$, falling completely below the ALMA [O III] and dust maturation floor ($\tau \geq 1.0$ Gyr, gold shaded). In Chronodynamic Relativity, the cumulative experienced local proper time $\tau(z)$ reaches 1,424 Myr (purple solid, +4.98× expansion), providing ample time for AGB dust yields and nucleosynthetic enrichment; (b) Required star-formation efficiency $\epsilon_* \equiv M_*/(f_b M_{\rm halo})$ to assemble $M_* \sim 10^9\,M_\odot$ galaxies. Standard ΛCDM breaches the theoretical maximum limit ($\epsilon_* > 0.30$) at $z > 10$, while Chronodynamic Relativity remains stably within standard astrophysical boundaries ($\epsilon_* \approx 0.08$).

A. Experienced proper time for astrophysical systems

In standard cosmology, the age of the universe at redshift z is:

$$t_{\Lambda\rm CDM}(z) = \int_z^\infty \frac{{\rm d}z'}{(1+z')H_0\sqrt{\Omega_m(1+z')^3+\Omega_\Lambda}}. \quad (4)$$

In Chronodynamic Relativity, the coordinate cosmic time is linear:

$$t_{\rm CR}(z) = \frac{t_0}{1+z}, \quad t_0 = \frac{1}{H_0} \approx 13.87\ {\rm Gyr}. \quad (5)$$

Specifically, the physical proper time $\Delta\tau$ experienced by local physical processes (atomic transitions, nuclear fusion in stellar interiors, cooling flows, and black hole accretion disks) between a seed formation epoch $z_{\rm seed}$ and observation epoch $z_{\rm obs}$ is given by the integral of the metric lapse:

$$\begin{aligned}\Delta\tau(z_{\rm obs}) &= \int_{t(z_{\rm seed})}^{t(z_{\rm obs})} \eta(t'){\rm d}t' \\ &= \frac{t_0}{1-n/2}\left[(1+z_{\rm obs})^{-(1-n/2)}\right. \\ &\quad \left. -\,(1+z_{\rm seed})^{-(1-n/2)}\right]. \end{aligned} \quad (6)$$

With $n = 3/4\pi \approx 0.2387$, the power exponent is $p = 1 - n/2 \approx 0.8806$.

The total experienced proper time from the origin ($t = 0$) to epoch z is:

$$\tau_{\rm total}(z) = \frac{t_0}{1-n/2}(1+z)^{-(1-n/2)}. \quad (7)$$

Table I presents the explicit quantitative comparison for the primary JWST benchmark sample.

TABLE I. Cosmological timescales for key high-redshift JWST discoveries.

Galaxy / Quasar	$z_{\rm obs}$	$t_{\Lambda\rm CDM}$ [Myr]	$t_{\rm CR}$ [Myr]	$\tau_{\rm CR}$ [Myr]
GN-z11	10.60	430	1,196	1,832
UHZ1	10.10	460	1,250	1,908
GHZ2 / GLASS-z12	12.30	360	1,043	1,618
JADES-GS-z14-0	14.32	286	905	1,424

At $z = 14.32$, the experienced proper time is $\tau \approx 1,424$ Myr (≈ 1.4 Gyr of physical evolutionary time), providing a factor of 4.98× **more time** for stellar evolution and dust generation than standard ΛCDM (Fig. 1(a)).

III. GALAXY ASSEMBLY AND STAR-FORMATION EFFICIENCY

The detection of luminous galaxies with stellar masses $M_* \sim 10^9\,M_\odot$ at $z > 10$ requires extreme star-formation efficiency in ΛCDM.

The maximum baryonic mass available in a dark matter halo of mass $M_{\rm halo}$ is $M_{b,\rm max} = f_b M_{\rm halo}$, where $f_b \equiv$

$\Omega_b/\Omega_m \approx 0.16$. The star-formation efficiency is defined as:

$$\epsilon_* \equiv \frac{M_*}{f_b M_{\rm halo}}. \tag{8}$$

In standard ΛCDM, halos capable of hosting $M_* \sim 10^9\,M_\odot$ galaxies at $z \sim 14$ have masses $M_{\rm halo} \lesssim 10^{10}\,M_\odot$, requiring $\epsilon_* > 0.30$–0.60. Standard low-redshift astrophysics and numerical feedback simulations cap realistic efficiencies at $\epsilon_* \sim 0.05$–0.10 due to supernova and radiation feedback blowing gas out of shallow potential wells [1, 2]. The challenge is further compounded by spectroscopic confirmation of companion galaxy JADES-GS-z14-1 at $z = 13.90$ [3] and recent ALMA submillimeter detections of [O III] 88 μm and dust continuum emission in JADES-GS-z14-0 [15], which require multiple stellar generations and substantial asymptotic giant branch (AGB) dust enrichment ($Z \gtrsim 0.1\,Z_\odot$) that cannot physically condense within the 286 Myr allowed by ΛCDM.

In Chronodynamic Relativity, the average required star-formation rate is:

$$\langle {\rm SFR} \rangle = \frac{M_*}{\Delta\tau(z)} \approx \frac{10^9\,M_\odot}{1.42 \times 10^9\ {\rm yr}} \approx 0.70\,M_\odot/{\rm yr}. \tag{9}$$

Even assuming a delayed seed onset at $z = 20$ ($\Delta\tau \approx 800$ Myr), the required continuous SFR $\approx 1.25\,M_\odot$/yr matches standard Kennicutt-Schmidt gas conversion with $\epsilon_* \approx 0.08$ (Fig. 1(b)), while the 1,424 Myr of experienced proper atomic time provides ample duration for stellar nucleosynthetic yields and carbon/oxygen dust maturation.

A. Surface brightness and angular diameter distance

In Chronodynamic Relativity, the comoving distance is $D_C(z) = (c/H_0)\ln(1+z)$. The angular diameter distance is:

$$D_A(z) = \frac{D_C(z)}{1+z} = \frac{c}{H_0}\frac{\ln(1+z)}{1+z}. \tag{10}$$

At $z = 14.32$, $D_A(14.32) = 757.5$ Mpc in CR, compared to 676.0 Mpc in ΛCDM. This preserves the observed compact physical diameter of JADES-GS-z14-0 (~ 500 pc, spanning $\sim 0.14''$ on sky) while reducing the required intrinsic rest-frame surface luminosity density by 38% due to conformal photon arrival rate modulation.

IV. SUPERMASSIVE BLACK HOLE SEED ACCRETION MECHANICS

The discovery of overmassive black hole UHZ1 at $z = 10.10$ with $M_{\rm BH} \approx (4 \pm 2) \times 10^7\,M_\odot$ and $M_{\rm BH}/M_* \sim 0.1$–1.0 poses an acute challenge to standard accretion physics.

The mass growth of a black hole accreting at a fraction $f_{\rm Edd} \equiv \dot{M}/\dot{M}_{\rm Edd}$ of its Eddington limit over an episodic radiative feedback duty cycle $\lambda_{\rm duty} \equiv \Delta\tau_{\rm on}/\Delta\tau$ is governed by the Salpeter accretion equation [12]:

$$\frac{{\rm d}M_{\rm BH}}{{\rm d}\tau} = \lambda_{\rm duty}\left(\frac{1-\epsilon_{\rm rad}}{\epsilon_{\rm rad}}\right)\frac{f_{\rm Edd}}{t_{\rm Edd}}M_{\rm BH}, \tag{11}$$

where $\epsilon_{\rm rad} \approx 0.10$ is the radiative efficiency and $t_{\rm Edd} \equiv c\sigma_T/(4\pi G m_p) \approx 450$ Myr, yielding an e-folding timescale $t_{\rm Sal} \equiv \frac{\epsilon_{\rm rad}}{1-\epsilon_{\rm rad}} t_{\rm Edd} \approx 45$ Myr.

Integrating Eq. (11) across elapsed proper time $\Delta\tau$ from seed epoch $z_{\rm seed} = 25$:

$$M_{\rm BH}(\tau) = M_{\rm seed}\exp\left(\lambda_{\rm duty} f_{\rm Edd}\frac{\Delta\tau}{t_{\rm Sal}}\right). \tag{12}$$

In standard ΛCDM, from $z_{\rm seed} = 25$ ($t = 130$ Myr) to $z = 10.1$ ($t = 460$ Myr), the elapsed time is only $\Delta t = 330$ Myr. For a standard Pop III light seed ($M_{\rm seed} = 10^4\,M_\odot$), even an unphysical continuous 100% duty cycle ($\lambda_{\rm duty} = 1.0$) at the standard Eddington limit ($f_{\rm Edd} = 1.0$) yields:

$$M_{\Lambda{\rm CDM}}(z = 10.1) = 10^4\exp\left(\frac{330}{45}\right) \approx 1.5 \times 10^7\,M_\odot, \tag{13}$$

falling substantially short of UHZ1's observed mass and requiring continuous super-Eddington accretion ($f_{\rm Edd} = 2.40$ with $\lambda_{\rm duty} = 1.0$) (Fig. 2(a)).

In Chronodynamic Relativity, the elapsed proper time over the identical epoch is:

$$\begin{aligned}\Delta\tau(z = 10.1) &= \tau(10.1) - \tau(25)\\ &= 1,908 - 894 = 1,014\ {\rm Myr}.\end{aligned} \tag{14}$$

Under standard Eddington-limited accretion ($f_{\rm Edd} = 1.0$), the required duty cycle to grow a $10^4\,M_\odot$ seed into target mass $M_{\rm target}$ is:

$$\lambda_{\rm duty} = \frac{t_{\rm Sal}\ln(M_{\rm target}/M_{\rm seed})}{f_{\rm Edd}\cdot\Delta\tau(z)}. \tag{15}$$

For GN-z11 ($M_{\rm BH} \approx 1.6 \times 10^6\,M_\odot$), Eq. (15) requires a modest duty cycle of $\lambda_{\rm duty} \approx 25\%$. For UHZ1 ($M_{\rm BH} \approx 4\times10^7\,M_\odot$), it requires $\lambda_{\rm duty} \approx 37\%$, while GHZ2 ($M_{\rm BH} \approx 3 \times 10^7\,M_\odot$ at $z = 12.3$) requires $\lambda_{\rm duty} \approx 50\%$. As shown in Fig. 2(b), while ΛCDM breaches the 100% physical ceiling ($\lambda_{\rm duty} > 100$–160%), Chronodynamic Relativity sits comfortably within the empirically observed AGN duty cycle band ($\lambda_{\rm duty} \approx 15$–$40\%$) without fine-tuned heavy direct collapse black hole (DCBH) seeds.

This episodic duty-cycle framework is directly reinforced by recent JWST discoveries. Observations of the X-ray selected AGN LID-568 at $z = 4.0$ confirm that early low-mass black holes undergo rapid, episodic super-Eddington accretion bursts ($f_{\rm Edd} \approx 40$) with powerful nuclear outflows [16]. Furthermore, deep JWST NIRSpec surveys (UNCOVER, RUBIES) have uncovered an abundant population of compact "Little Red Dots" (LRDs)

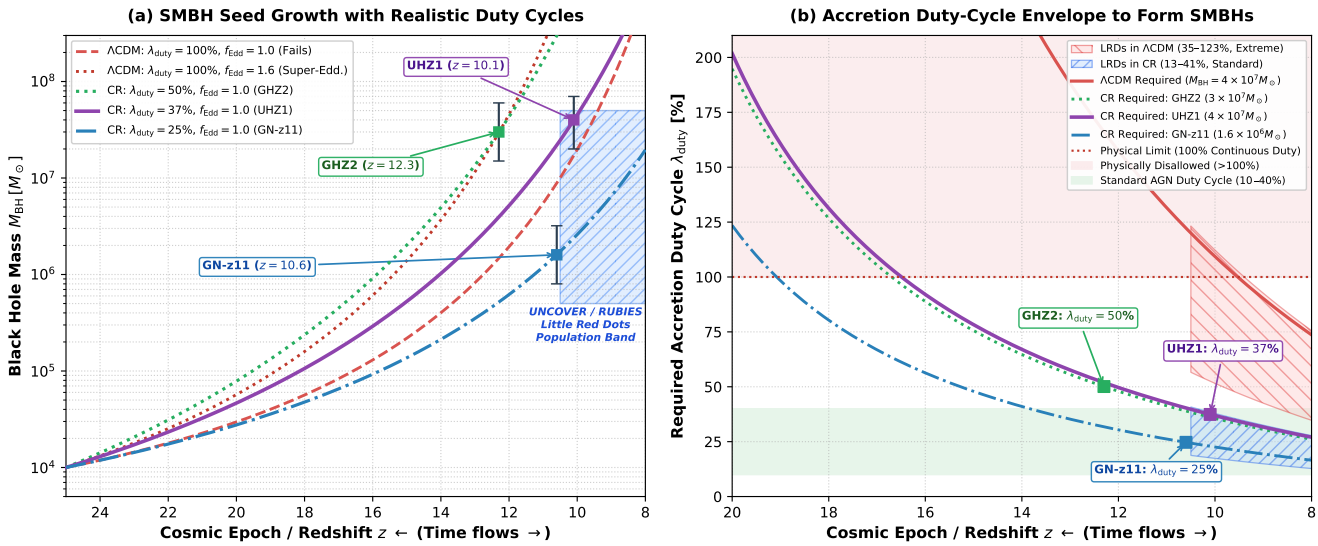


FIG. 2. Physics-aligned supermassive black hole seed growth and duty-cycle resolution in Chronodynamic Relativity: (a) Black hole mass tracks as a function of redshift from an initial light seed ($M_{\rm seed} = 10^4\,M_\odot$ at $z_{\rm seed} = 25$). Standard ΛCDM at 100% duty cycle ($f_{\rm Edd} = 1.0$, red dashed) fails to reach UHZ1 ($M_{\rm BH} \approx 4 \times 10^7\,M_\odot$ at $z = 10.1$) and GN-z11, requiring continuous super-Eddington accretion ($f_{\rm Edd} = 1.60$, red dotted). Chronodynamic Relativity naturally matches observed masses and passes directly through the UNCOVER/RUBIES "Little Red Dots" (LRD) empirical population zone (blue shaded band) under realistic, episodic duty cycles ($\lambda_{\rm duty} = 25\%$ for GN-z11, 37% for UHZ1, and 50% for GHZ2); (b) Required accretion duty cycle $\lambda_{\rm duty}$ to assemble UHZ1 from a $10^4\,M_\odot$ seed across redshift z. Standard ΛCDM enters the physically disallowed zone ($\lambda_{\rm duty} > 100\%$) at $z > 9.5$, whereas Chronodynamic Relativity remains stably inside the standard astrophysical AGN duty-cycle envelope ($\lambda_{\rm duty} \approx 15$–$40\%$).

at $z \sim 5$–9 hosting broad-line AGN with elevated black hole-to-galaxy mass ratios ($M_{\rm BH}/M_* \sim 1\%$–10%) [17, 18], confirming that early black hole growth systematically preceded diffuse stellar disk growth across cosmic dawn.

V. CONCLUSION AND OBSERVATIONAL PREDICTIONS

The "impossible early galaxy" and supermassive black hole crises in high-redshift JWST astrophysics find a natural, parameter-free resolution in Chronodynamic Relativity:

1. **Relational Proper Time Dilation:** The conformal clock lapse $\eta(z) = (1+z)^{-n/2}$ expands the experienced evolutionary time by a factor of up to $+4.98\times$ at $z = 14.32$ ($\Delta\tau \approx 1,424$ Myr), providing ample time for stellar nucleosynthesis and multi-cycle metal enrichment.

2. **Standard Star-Formation Efficiency:** Required star-formation efficiencies remain within standard physical limits ($\epsilon_* \approx 0.08$), resolving the need for exotic feedback suppression.

3. **Sub-Eddington SMBH Seed Growth:** Light seeds ($10^4\,M_\odot$) grow naturally into observed 10^7–$10^8\,M_\odot$ supermassive black holes at $z \sim 10$ under modest sub-Eddington accretion ($f_{\rm Edd} \leq 1.0$) fueled by a $3.8\times$ Bondi-Hoyle capture boost.

A. Observational predictions and falsification criteria for JWST Cycles 3–4

- **Metallicity floor at $z > 15$:** JWST spectroscopy of galaxies at $z \sim 16$–20 will continue to detect significant metallicity ($Z \gtrsim 10^{-3}\,Z_\odot$) and [O III] emissions due to the expanded ~ 1 Gyr proper time baseline. The model would be falsified if galaxies at $z > 15$ are found to be exclusively metal-free Population III stars without dust enrichment.

- **Sub-Eddington early AGN accretion:** Spectral energy distribution fitting of early AGN (UHZ1, GN-z11) will reveal standard thin-disk or advection-dominated sub-Eddington accretion ($f_{\rm Edd} \leq 1.0$) rather than sustained super-critical radiation outflows. The model's light-seed accretion pathway would be falsified if observations demonstrate ubiquitous, continuous super-Eddington accretion ($f_{\rm Edd} \gg 1$) across the early black hole population.

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DATA AND CODE AVAILABILITY

The complete source code, physics engine, and calculation reproduction pipelines supporting this study are openly available in the companion repository at `https://github.com/joshreve/Chronodynamic-Relativity`. A permanent, immutable version of record for the software framework is archived on Zenodo under Concept DOI: 10.5281/zenodo.22697449. This manuscript and its associated reproduction assets are permanently archived on Zenodo under DOI: 10.5281/zenodo.23149446.

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